

G.R.I.P

Glove Rejection and Inspection Process

➔ Problem & Solution

Problem: Right-hand and unclear/crumpled gloves appear on a left-glove conveyor lane. Manual inspection is slow and inconsistent.

Solution: A parallel SCARA robot removes rejected gloves automatically in real time.

➔ Scope & Pipeline

Prototype: One-lane system - left gloves pass, right & unclear rejected.

Pipeline:

Camera → YOLO Detection → Decision Logic → PC/UI → STM32H7 → ODrive → Parallel SCARA → Pneumatic End Effector → Reject Bin

- ✓ Real-time detection
- ✓ Glove classification
- ✓ Automated rejection
- ✓ Factory-ready

